

A DEEP LEARNING LAB (AIM 3230) SYNOPSIS on

Context-Aware Harmful/Violent Sentiment Detection in Twitter Threads

Submitted to Manipal University Jaipur

Towards the partial fulfillment for the Award of the Degree of

**B. Tech Computer Science and Engineering (Artificial Intelligence and
Machine Learning**

2023-2027

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1. Introduction

Twitter (X) serves as a major platform for public discussions through short posts and threaded conversations. While Twitter threads encourage engagement and opinion sharing, they are also frequently used to propagate harmful content such as violent expressions, hate speech, racism, and abusive language. Due to the scale and speed of such interactions, manual moderation becomes infeasible.

Recent advancements in Deep Learning and Natural Language Processing (NLP) have enabled automated systems to analyze contextual sentiment in text. This project proposes a deep learning-based approach to analyze Twitter threads and detect harmful or violent sentiment, thereby demonstrating the practical application of neural networks in real-world social media analysis.

2. Problem Statement

Conventional sentiment analysis approaches classify text into generic categories such as positive or negative sentiment, which fails to capture nuanced harmful behavior present in Twitter threads. Additionally, tweets contain informal language, abbreviations, hashtags, sarcasm, and contextual dependencies across multiple replies.

The challenge addressed in this project is to design and train a deep learning model capable of understanding sequential and contextual information within Twitter threads to accurately detect harmful or violent sentiment.

3. Objectives

- To implement deep learning models from scratch using feedforward neural networks
- To apply gradient descent and backpropagation for training
- To utilize advanced optimizers such as Adam and RMSprop
- To analyze Twitter thread data using CNN and RNN architectures
- To apply regularization techniques to improve generalization
- To evaluate model performance using real test examples

4. Proposed Methodology

4.1 Overall Architecture

The proposed system is designed to detect harmful sentiment in Twitter threads using deep learning techniques. The implementation is carried out using Python and the TensorFlow–Keras framework. Supporting libraries such as Pandas, NumPy, and Scikit-learn are used for data preprocessing, numerical computation, and evaluation.

The architecture follows a structured pipeline:

1. Data Collection
2. Text Preprocessing
3. Tokenization and Sequence Encoding
4. Embedding Representation
5. Recurrent Neural Network Model
6. Dense Output Layer
7. Model Evaluation

4.2. Text Preprocessing

Twitter threads (including the main tweet and associated replies) are merged into a single textual sequence to preserve conversational context.

The preprocessing stage includes:

- Conversion of text to lowercase
- Removal of URLs, mentions, hashtags, and special characters
- Removal of punctuation and irrelevant symbols
- Tokenization into word sequences
- Conversion of words into integer indices
- Padding or truncating sequences to a fixed maximum length

Padding ensures uniform input dimensions for batch training in neural networks.

4.3 Feature Representation

After preprocessing, the text is tokenized and each word is assigned a unique integer index based on the training vocabulary. The sequences are padded to a fixed length before being fed into the model.

These integer sequences are passed to a trainable Embedding layer, which converts each word index into a dense vector of fixed dimension. The embedding layer functions as a learnable lookup matrix where:

- Each word in the vocabulary has a corresponding vector
- Word vectors are updated during backpropagation
- Words used in similar contexts gradually develop similar representations

Unlike pretrained embeddings (e.g., Word2Vec, GloVe), the embeddings are learned directly from the Twitter dataset, making them task-specific and better suited to informal language.

The resulting embedding output is then passed to the LSTM/GRU layer for contextual sequence modeling.

4.4 Recurrent Neural Network Architecture

Since Twitter threads are sequential in nature, a **Recurrent Neural Network (RNN)** is employed to capture contextual dependencies between words.

The following architectures are considered:

- **Long Short-Term Memory (LSTM)**
- **Gated Recurrent Unit (GRU)**

The number of recurrent units is treated as a tunable hyperparameter (e.g., 64–128 units), selected based on validation performance. This flexibility allows the model to balance representational capacity and computational efficiency.

4.5 Regularization

To prevent overfitting, a **Dropout layer** is applied after the recurrent layer.

The dropout rate is treated as a tunable parameter (typically in the range of 0.3–0.5) and selected experimentally. This helps improve generalization to unseen Twitter threads.

Optional techniques such as early stopping may also be applied during training.

4.6 Output Layer

A fully connected **Dense layer** with a **sigmoid activation function** is used for binary classification.

- Output range: 0 to 1
- Threshold (commonly 0.5) used to classify:
 - Harmful
 - Non-harmful

4.7 Model Training

The model is trained using:

- **Loss Function:** Binary Cross-Entropy
- **Optimizer:** Adam (adaptive learning rate optimization)
- **Batch Size and Epochs:** Selected experimentally

Backpropagation is used to update model weights iteratively by minimizing classification error.

Hyperparameters such as learning rate, number of units, and dropout rate are tuned based on validation accuracy and loss trends.

4.8 Prediction and Evaluation

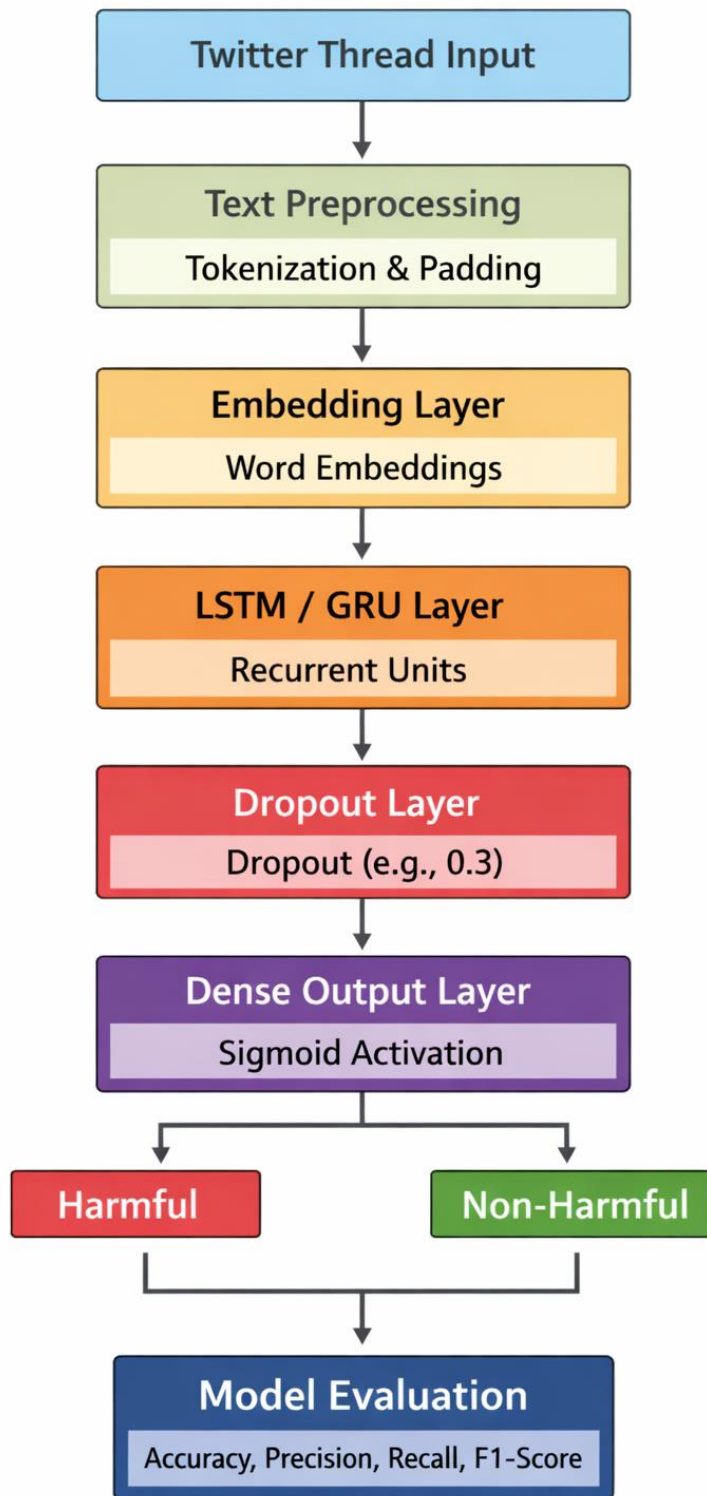
After training, the model is evaluated using unseen test data.

Evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix (optional)

Since harmful sentiment detection may involve class imbalance, F1-score is considered an important metric for balanced performance assessment.

Performance on validation and test datasets is analyzed to ensure that the model generalizes well beyond the training data.



5. Applications

- Detection of harmful or abusive content on social media platforms
- Automated monitoring of online discussions
- Support systems for content moderation and policy enforcement

6. Conclusion

This project presents a deep learning-based approach for detecting harmful sentiment in Twitter threads. By combining NLP preprocessing techniques with neural network models, the system can analyze contextual information present in threaded conversations. The project provides hands-on experience with core deep learning concepts and demonstrates their application to a relevant and practical problem in social media analysis.